

CS:5980

Foundations of Embedded Systems

Safety Requirements

Part III

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Requirements-based Design

Systematic approach to design of systems

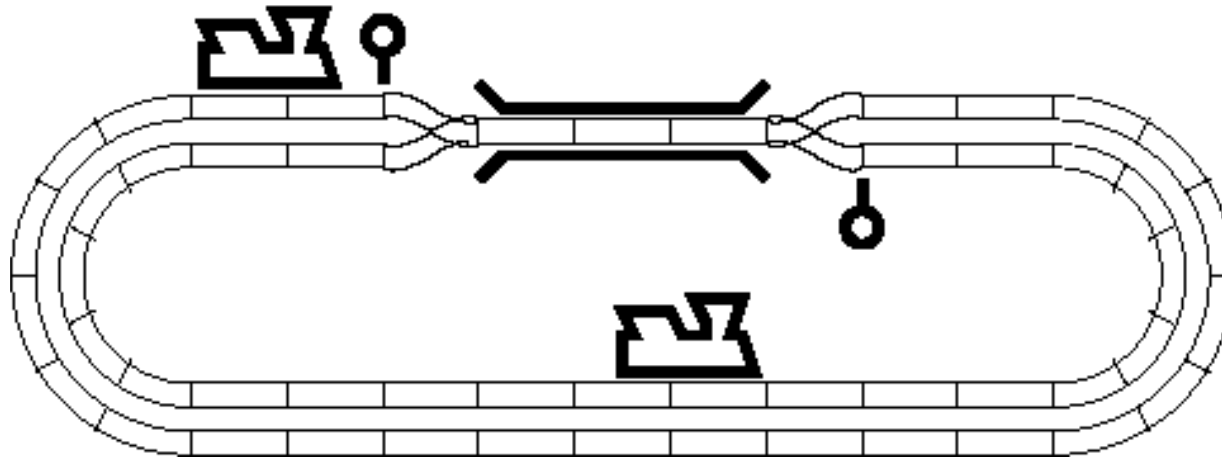
Given:

- Input/output interface of system/component C to be designed
- Model E of the environment
- Safety properties $P_1, \dots, P_n$ of the composite system

Design problem:

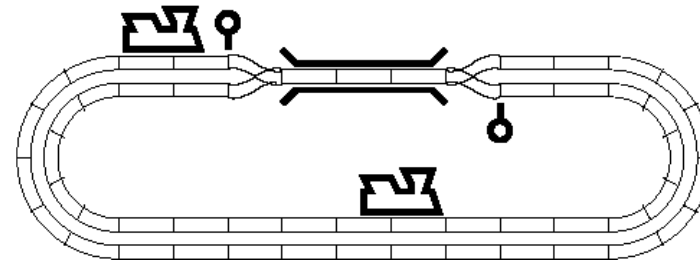
- Fill in details of C (state variables, initialization, and update) so that $P_1, \dots, P_n$ are invariant for $C \parallel E$

Railroad Controller Example



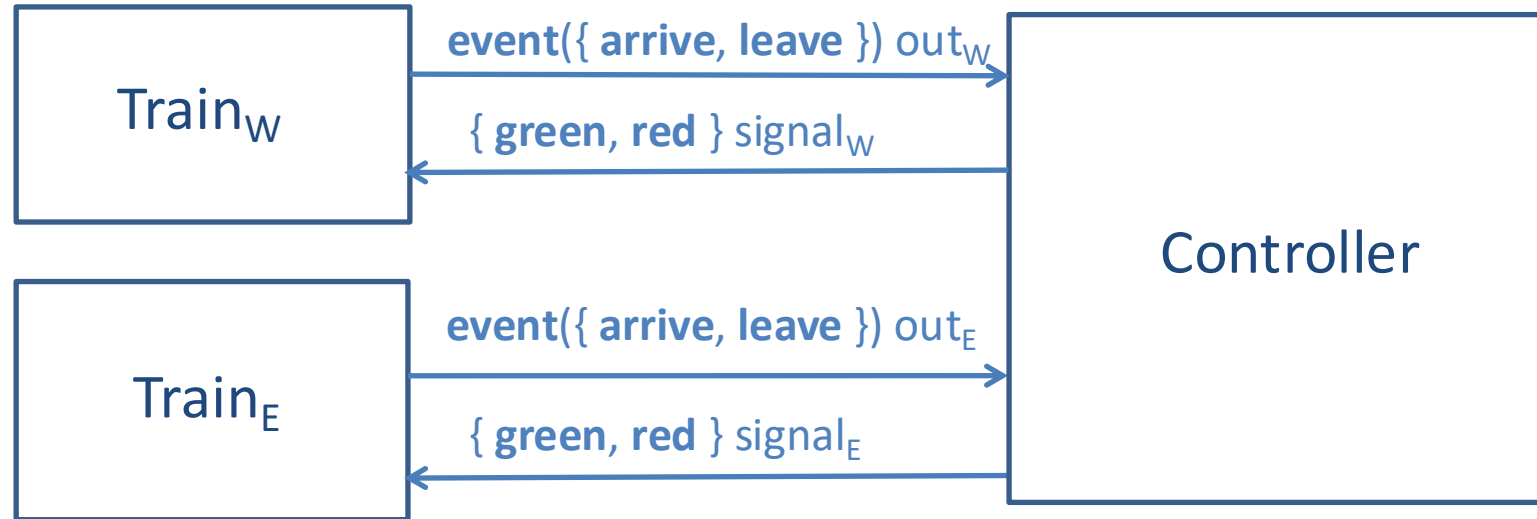
Goal: design controller for the two signals at the bridge

Train Model



- Each train is initially **away** from bridge
- Train can be **away** for an arbitrarily long period
- When a train gets close, it communicates with the traffic controller via an **arrive** event, and moves to a **wait** state
- When on arrival to the bridge, train monitors the signal there:
 - If the signal is **green**, it enters the **bridge**
 - If the signal is **red**, it continues to wait
- A train can stay on **bridge** for an undetermined duration (which is not directly under the control of the traffic controller)
- When the train leaves the bridge, it communicates with the controller by issuing a **leave** event and returns to **away** state
- This behavior repeats: an away train may again request entry as above
- The two trains have symmetric behavior (each relying on its corresponding signal)

Controller Design Problem

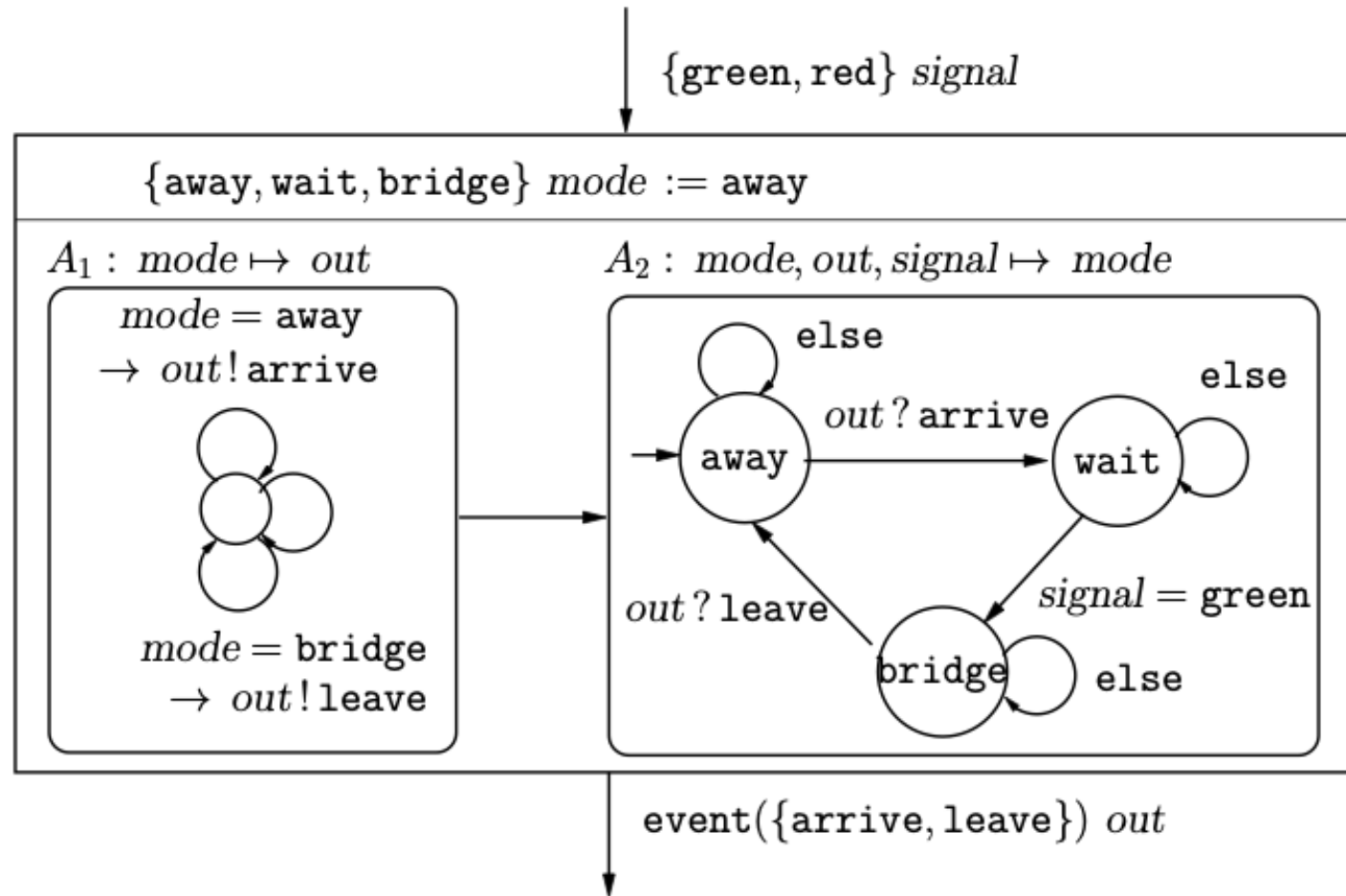


Safety Requirement: Trains should not be on bridge simultaneously

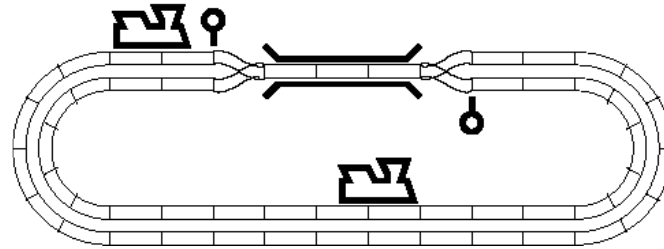
Formally, the following should be an **invariant** of the system:

$$\neg(\text{mode}_W = \text{bridge} \wedge \text{mode}_E = \text{bridge})$$

Synchronous Component Train

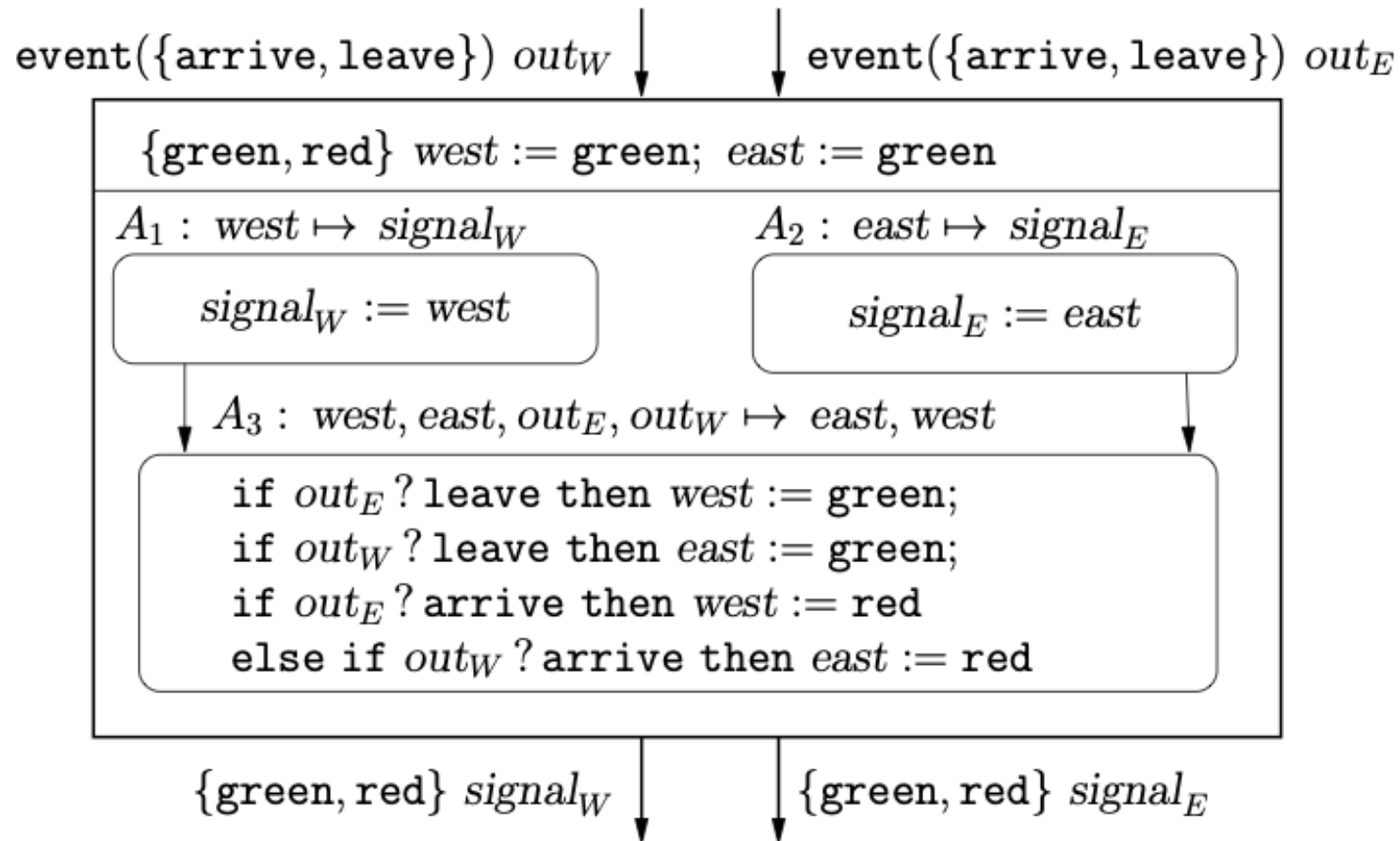


First Attempt at Controller Design



- Controller maintains state variables **east**, **west** to track the state of each signal
- Both state variables are initially **green**
- The output for the signals is based on the corresponding state vars
- When a train arrives, update the opposite signal var to **red** to block the other train from entering the bridge
- When a train leaves, reset the opposite signal var to **green**
- What happens if both trains arrive simultaneously?
 - Give priority to east train: set **west** signal var to **red**

Synchronous Component Controller1



Controller

Train W

Train E

west

east

mode_W

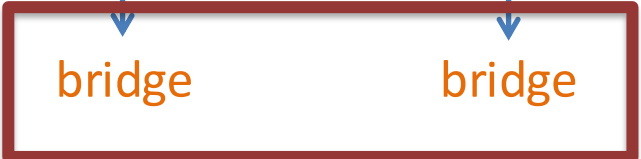
mode_E

green
↓
red
↓ red!
red
↓ red!
green
↓ green!
red
↓ red!
red

green
↓
green
↓ green!
green
↓
green
↓
green
↓ green!
green

away
↓ arrive!
wait
↓
wait
↓
wait
↓
bridge
↓
bridge

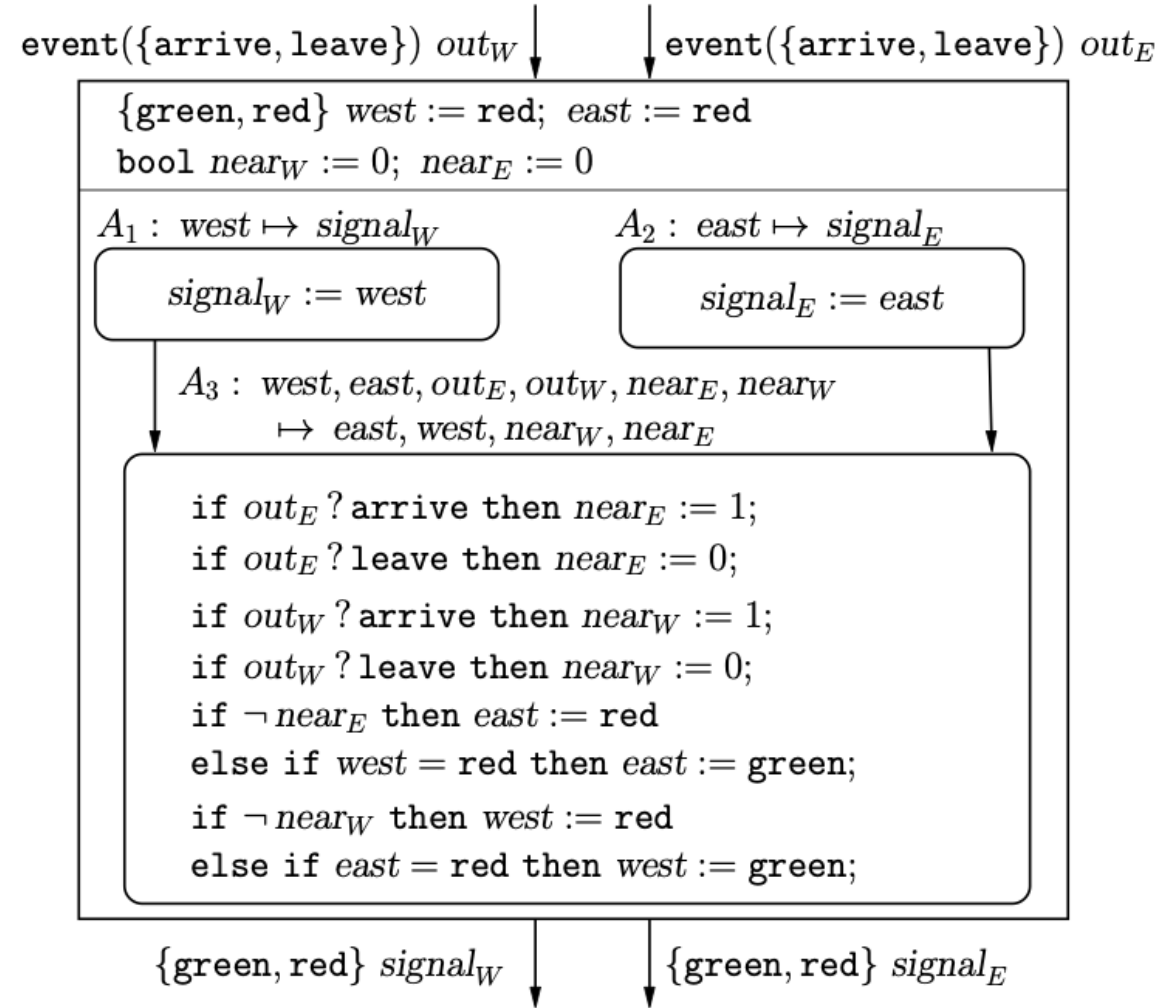
away
↓ arrive!
wait
↓
bridge
↓ leave!
away
↓ arrive!
wait
↓
bridge



Second Attempt at Controller Design

- What went wrong the first time?
 - Controller did not keep track whether a train was waiting at each entrance
- Boolean variable near_w remembers if the west train wants to use the bridge
 - Initially 0
 - When the west train issues **arrive**, changed to 1
 - When the west train issues **leave**, reset back to 0
- Controller invariant: $\text{mode}_w = \text{away} \Leftrightarrow \text{near}_w = 0$
- Variable near_e is defined symmetrically
- Let's also start with both signals **red**
- A signal is changed to **green** if the corresponding train is near and the other signal is **red**; it is changed back to **red** when train is away
- Need still to resolve simultaneous arrivals by preferring one train

Second Attempt at Controller Design



Properties of Controller2

- $\text{RailRoadSystem2} = \text{Controller2} \parallel \text{Train}_W \parallel \text{Train}_E$ satisfies the safety property
 $\neg(\text{mode}_W = \text{bridge} \wedge \text{mode}_E = \text{bridge})$
- What about some additional properties?
 1. *If the west train is waiting, then west signal will eventually become green*
 2. *If the west train is waiting, other train should not be allowed on bridge more than once*
- Requirement 1 is a **liveness** requirement (see Chap. 5 of text)
- Requirement 2 is a **safety** requirement
 - Its violation can be demonstrated by a (finite) execution in which east train enters, leaves, and enters again while west train keeps waiting with its signal red
 - But it cannot be encoded as an invariant on system state vars!

Safety Monitor

Monitor M for a system observes its inputs/outputs, and enters an error state if undesirable behavior is detected

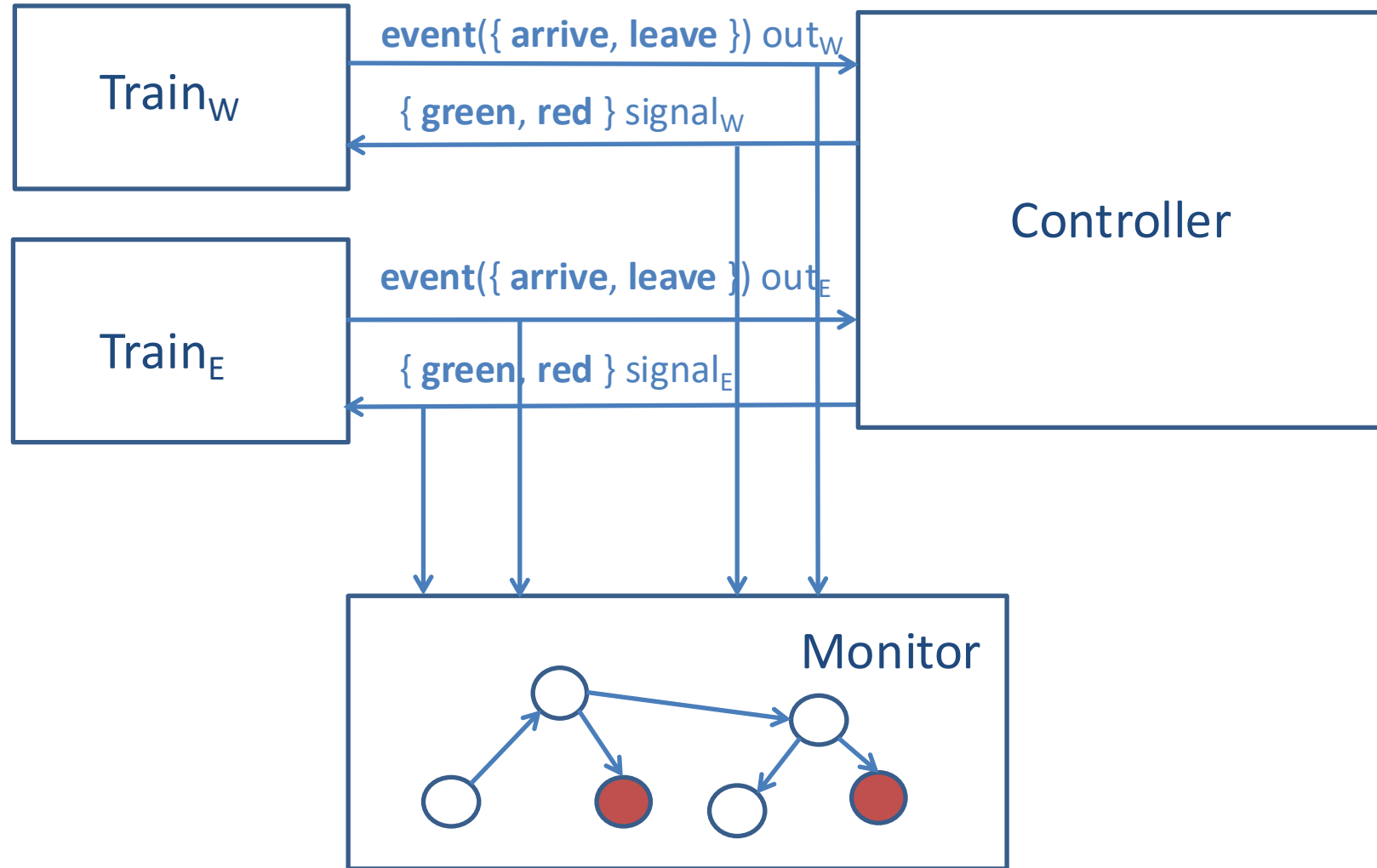
Monitor M is specified as extended state machine

1. The set of **input variables** of $M \subseteq \text{input/output variables}$ of system being monitored
2. An output of M cannot be an input to system
(monitor does not influence what the system does)
3. A subset F of of state-machine **modes** are considered as *bad*

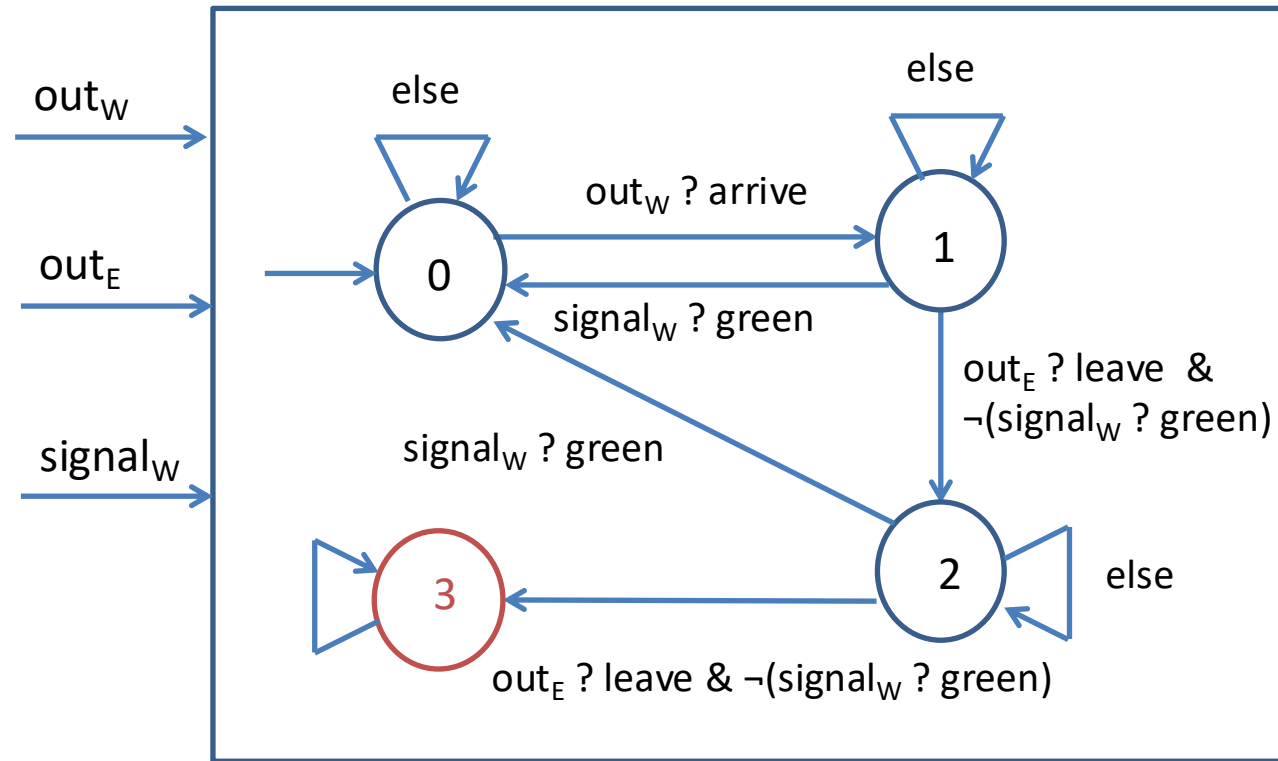
Undesirable behavior: An execution that leads monitor state to F

Safety verification: **Check** whether $M.\text{mode} \notin F$ is an **invariant** of $C \parallel M$

Safety Monitors



Monitor to check fairness for railroad



Error execution: As west train waits, east train is allowed on bridge twice

Exercise: Leader Election

Suppose we want to check that at most one of the nodes declares itself to be the leader

1. Design a monitor M

- Input variables: $\{\text{undecided}, \text{leader}, \text{follower}\} \text{status}_n$, for each node n
- M should enter error state iff for two distinct nodes m and n
 - there exists a round r_1 in which $\text{status}_m = \text{leader}$ and
 - there exists a round r_2 in which $\text{status}_n = \text{leader}$

2. Consider the requirement: eventually $\text{status}_n \neq \text{undecided}$

Why can't we design a monitor that enters an error state if this requirement is violated?

Credits

Notes based on Chapter 3 of

Principles of Cyber-Physical Systems

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